Decimal Place-Value Builder

TEACHER KEY

5th Grade Math · Interactive Math · Explore & Reflect

Get started

Build numbers digit by digit in a place-value chart, multiply and divide by 10, and see expanded form, word form and base-ten blocks change. Open the tool and use it to answer each question.

Explore

1. Build 3.47. Write it in expanded form.

$3 \times 1 + 4 \times \frac{1}{10} + 7 \times \frac{1}{100}$

2. Build 3.47 and press × 10. What number do you get?

34.7

3. Start again with 3.47 and press ÷ 10. What number do you get?

0.347

4. In 5.55, the first 5 is worth how many times the second 5?

10 times

5. Build 0.125. Write it in word form.

one hundred twenty-five thousandths

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Explore more

6. Which is greater: 0.6 or 0.58? Use the chart to explain.

0.6, because 6 tenths > 5 tenths

7. Build 2.3 with blocks. How many rods are there?

3 rods (3 tenths)

Reflect

8. Reflect: Why do the digits move left when you multiply by 10?

Each place is 10 times the place to its right, so each digit becomes worth 10 times as much.

Teacher notes

Accept equivalent forms (for example 1/2 = 2/4 = 0.5) and any correct example where several answers are possible. Have students explain using the tool’s picture.